MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG4DSEBOTA05** Course Title: **Bio-Instrumentation and Quality Analysis 2**

Time: 2 Hrs **Maximum Marks: 70**

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part **A**:
Answer in One or Two sentences
Answer 10 questions
Each question carries 2 mark

1	What is RPM?	Remember	1
2	What is freeze-thawing?	Remember	1
3	Write any two uses of refrigerated centrifuge	Remember	1
4	What is the use of filter aid?	Remember	1
5	How will you calculate Rf value	Understand	2
6	What do you mean by adsorption chromatography?	Remember	2
7	What is the use of gel documentation system?	Remember	3
8	What you mean by Isoelectric focusing?	Remember	3
9	What is the use of stage micrometer	Remember	3
10	What is camera lucida	Remember	3
11	Describe any two applications of Scintillation counter	Remember	4
12	Explain hemocytometer	Understand	4

Part **B**:
Short Answer Type Questions
Answer 8 Questions
Each question carries 5 marks

13	Describe the principle and working of tangential filtration	Remember	1
14	Describe the principle, working and application of preparative ultracentrifuge	Remember	1
15	Describe the use of detergents and enzymes in cell disruption	Remember	1
16	Explain the applications of chromatography in pharmaceuticals and environmental science	Apply	2

17	Explain the principle and working of GC	Understand	2
18	Explain the preparation and applications of agarose gel electrophoresis	Understand	3
19	Explain the preparation and applications of Native PAGE	Understand	3
20	List out the applications of cell counting techniques in cell culture	Analyze	4
21	List out the applications of cell counting techniques in Immunophenotyping	Analyze	4
22	List out the applications of cell counting techniques in single cell analysis	Analyze	4

Part **C**:
Essay Type Questions
Answer any one question
Each Question carries 10 marks

23	With diagram explain the principle and working of Size exclusion chromatography. Add a note on its applications	Apply	2
24	Analyze the principle, working and applications of Southern blot	Analyze	3